IS222 2023 - Midterm Test on Database Management Systems

Database Management Systems (The University of the South Pacific)

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Student Name: _____

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Mid-Semester Test

IS222 – Database Management Systems

Semester I, 2023

Mode: Face-to-Face & Online Students

Date: Thursday, 13th April June 2023

Time: 3.00 –3.55 PM (FJ Time)

Weight: 25%

You have 5 minutes to read this paper,

Followed by 50 minutes to complete this paper.

All Questions Are Compulsory

This Test 1 will focus on the following learning outcomes for this course:

CLO1. Explain the fundamental concepts of various database management systems. CLO3. Construct a data model for a relational database management system.

Time allowed: 55 minutes only

Total Marks: 40

Total Pages: 7

Weight: 25%

INSTRUCTIONS:

1. This exam is worth 25% of your coursework.
2. ALL questions are compulsory.
3. Write your Name & ID # on all sheets in this booklet.
4. Zero for any Plagiarism case.

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SECTION A – [10 Marks]

1. Explain what data modeling is? [2 marks]

2. Consider the following table: [Identifying anomalies in relations]

Project Number	Project Name	Employee Number	Employee Name	Department	Hours Spent
1	Apollo	1001	Smith	Engineering	127
1	Apollo	1003	Jones	Accounts	45
2	Mercury	1002	Stewart	Marketing	70
3	Venus	1001	Smith	Engineering	21
3	Venus	1006	Brown	Engineering	124

Identify the potential problems (at least 2) that may arise if the data was kept in this relation.

[2 marks]

3. Use the following tables to answer question 3.

Table name: STUDENT

STU_CODE	PROF_CODE
100278	
128569	2
512272	4
531235	2
531268	
553427	1

Table name: PROFESSOR

PROF_CODE	DEPT_CODE
1	2
2	6
3	6
4	4

i Perform the following Relations Algebra on the above two tables and show the result: **Natural Join** [3 marks]

Hint: You must redraw and arrange the table to complete the task.

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ii Perform the following Relations Algebra on the above two tables and show the result:
Left outer join [3 marks]

SECTION B – [30 Marks]

4. Use the following description to answer parts (i) & (ii) of this question

You have been hired to design a small database for a new company called StarkCorp. StarkCorp only needs a few details tracked in their database which deals with its customers, products, and invoices.
As far as invoices are concerned, your newly designed database needs to store its unique invoice ID, sale date, product quantity, total cost due, and delivery date. We will assume that the delivery address is exactly the same as the customer's address, so it's not recorded in the invoice. Similarly, the company's products need to be tracked which consist of the following information: the distinct product code (e.g. P101), description (e.g. Trek Suitcase) and quantity available (such as 24). Apart from the details above, the system must track the customer details. These include the customer name (e.g. Rick R. Wayne), email address, street address (e.g. 7 Derby Street), city, state and phone contact stored as a whole number. Assume that all customers and sales are local and not international.
More importantly, each customer has been issued one or more invoices whilst each invoice is issued to exactly one customer. Similarly, each invoice consists of only one product. Also, each product, like the Trek Suitcase, can generate multiple invoices. On some rare occasions, it is possible that a product doesn't generate any invoice.

Your database must fulfill these requirements, which is all that is needed for now.

i List the business rules that you have used to design your database [4 marks]

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ii Draw your Crow's Foot Entity Relationship Diagram (or ERD). Be sure to specify the entities, attributes, primary/foreign keys, relationship strengths/cardinalities etc. [8 marks]

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5. Differentiate between entity and instance. [1 mark]

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_____

_____

6. What is a database? Give a real life example where you have used a database. [1 mark]

_____

_____

_____

_____

Normalization: The following data was previously maintained in a spreadsheet. You are required to normalize the data in the time sheet from 1NF to 2NF to 3NF by answering questions i to iv.

Order Number	Date	Customer Number	Name	Address	Product Number	Description	Price	Quantity
1	5/11/2004	22	Alan Smith	London	A95	Jacket	55	4
					G17	Coat	120	8
					K10	Suit	90	5
2	19/11/2004	47	Jay Jones	Paris	G17	Coat	120	9
					D77	Shirt	35	20
3	27/11/2004	25	Sally West	Glasgow	E30	Tie	5	25
					D77	Shirt	35	4

Note: 1NF describes tabular format in which; all key attributes are defined; there are no repeating groups in the table, and; all attributes are dependent on the primary key.

i Convert the above data into a tabular format that satisfies 1NF. [4 marks]

Hint: You must redraw and arrange the table to complete the task.

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ii Create a dependency diagram clearly showing partial and/or transitive dependencies (if any). The identified primary key(s) (PK) must be clearly shown. Any assumptions made must be stated clearly. [4 marks]

iii Show the dependency diagrams for each table in 2NF. [4 marks]

iv Show the dependency diagrams for each table in 3NF. [4 marks]

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All The Best!
The End..

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